Fiscal Year:	FY 2023	**Task Last Updated:**	FY 06/14/2023
PI Name:	Weber, Richard Ph.D.		
Project Title:	The Origin of Fragility in High-Temperature Oxide Liquids		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rweber@matsdev.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	847-612-8597
Organization Name:	Materials Development Inc.		
PI Address 1:	3090 N Daniels Ct		
PI Address 2:			
PI Web Page:			
City:	Arlington Heights	**State:**	IL
Zip Code:	60004-7708	**Congressional District:**	8
Comments:	NOTE: Formerly with Containerless Research, Inc., until 2006.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Japanese Space Agency (JAXA) AO for Electrostatic Levitation Furnace (ELF) experiments on ISS
Start Date:	08/15/2019	**End Date:**	09/30/2024
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Sansoucie, Michael	**Contact Phone:**	256.544.5269
Contact Email:	michael.p.sansoucie@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 9/30/2024 per NSSC information (Ed., 3/5/24) NOTE: End date changed to 9/30/2023 per NSSC information (Ed., 6/20/23) NOTE: End date changed to 9/30/2022 per M. Sansoucie/MSFC (Ed., 1/28/21) June 2019: Note project had preliminary work performed Feb-August 2019		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K1288 ; 80NSSC19K0518		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project is the United States (US) portion of an international Japan Aerospace Exploration Agency (JAXA) program to investigate the properties and structure of molten and supercooled metal oxides using the Electrostatic Levitation Furnace (ELF) on the International Space Station (ISS) and ground-based synchrotron x-ray and neutron studies. Four prototype compositions have been levitated and melted on ISS. Measurements of thermophysical properties, density, thermal expansion coefficient, viscosity and surface tension were made over a wide temperature range. Samples, including glasses, were recovered for ground based analysis and comparison with materials made on Earth. The work underpins development of a materials property-structure data set that can help in development of new materials and in meeting requirements of material utilization and processing in space. Additional samples are being prepared for measurements on ISS in 2023. Work is being published to provide data to the broader scientific community. The project team brings together the skills and experience needed to implement a complex research program. Reference: [1] S. Kohara, et al., “The origin of fragility in high-temperature oxide liquids - towards fabrication of novel non-equilibrium oxides,” JAXA flight project funded in 2017.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Developing an advanced understanding of the properties and structure of liquids that can be transformed into glasses is a high value enterprise that benefits the $300B global glass materials industry. Specific advances expected from this research will impact development of functional glasses by providing high accuracy thermophysical property and atomic structure data that can be used to optimize models used to design new materials from first principles. The titanate liquids being studied have high value applications. The compositions also relate to lunar mare regolith that contains relatively large amounts of titanium oxide. The work has led to discovery a new type of glass with highly coordinated titanium ions. Work is/has been published and a US patent on an optical device glass was filed.
Task Progress:	This project is part of a collaborative international Japanese Space Agency (JAXA) / NASA program to investigate: (i) properties of liquid phase and supercooled metal oxides using the Electrostatic Levitation Furnace (ELF) on the International Space Station (ISS), and (ii) atomic structure using ground-based synchrotron and neutron studies. Samples were launched on SpaceX CRS-20 (SpaceX-20) in spring of 2020, on SpaceX-22 in June 2021, and on SpaceX-27 in March 2023. The research is helping to establish the basis for designing advanced materials from first principles. This approach will make more competitive products at lower cost. The titanium oxide-based compositions being studied are useful for optical device applications including medical lasers and high speed communications. These compositions also relate to lunar mare regolith that contains relatively large amounts of titanium oxide. The work has led to discovery of a new type of glass with highly coordinated titanium ions. Work is/has been published and a US patent on an optical device glass was filed.
Bibliography Type:	Description: (Last Updated: 06/06/2026)
Articles in Peer-reviewed Journals	Weber R, Wilke SK, Benmore CJ. "Containerless techniques for in-situ X-ray measurements on materials in extreme conditions." J. Phys. Soc. Jpn. 2022 Sep 15;91(9):091008. http://dx.doi.org/10.7566/JPSJ.91.091008 , Sep-2022